

Rishi Parekh

✉ therishiparekh@gmail.com | 📞 +91 95100 35006 | 🔗 LinkedIn | 🐙 GitHub | 📁 Leetcode

EDUCATION

Pandit Deendayal Energy University
B.TECH IN COMPUTER ENGINEERING

JULY 2023 – MAY 2027
CGPA: 8.35/10

EXPERIENCE

Web & Mobile Application Development Intern
INDIAN INSTITUTE OF TEACHER EDUCATION (IITE)

MAY 2026 – AUG 2026
GANDHINAGAR, INDIA

Key Skills: Node.js, Express, MongoDB, Redis, JWT, Firebase, React, Tailwind CSS, Flutter, Socket.io

- Built a full-stack University ERP for IITE using **Flutter**, **React**, **Node.js**, and **MongoDB**, serving Students, Faculty, and Administrators with attendance, timetable, announcements, and academic management modules.
- Engineered secure **JWT + RBAC** authentication with **Redis in-memory caching (3600s TTL)**, reducing database read load by **90–95%** and delivering sub-**10ms** response times for frequently accessed academic data.
- Improved API responsiveness by **~95%** using **BullMQ** background workers for attendance processing and report generation, ensuring non-blocking operations and a seamless user experience.
- Optimized MongoDB performance through **compound indexing**, atomic upserts, and Dockerized deployment with health monitoring, enabling up to **100× faster queries**, preventing duplicate attendance records, and improving scalability and system reliability.

Freelance Full-Stack Web Developer
SELF-EMPLOYED

JAN 2024 – PRESENT
GANDHINAGAR, INDIA

Key Skills: React.js, Flutter, Tailwind CSS, GSAP, Node.js, Express.js, REST APIs, Redis, MongoDB, Firebase, MySQL, Git, GitHub, Vercel, Render, Razorpay

- Independently designed and shipped 4 production client websites – Three O’Clock Cafe, Vellarossa, Chef Choice Consultancy, and Strata Q – spanning hospitality and B2B consulting.
- Engineered scroll-linked animation systems using GSAP ScrollTrigger, Framer Motion, and Lenis, synchronizing custom page-transition engines across React/Next.js SPAs.
- Owned the full engineering lifecycle – architecture, deployment, domain configuration, and client-facing updates – generating Rs. 60,000+ in revenue across 4 delivered projects.
- Solved recurring technical challenges independently – animation memory leaks, scroll/router race conditions, and hybrid video rendering logic – across every build.

PROJECTS

Arihant Store | MERN, NEXT.JS, REDIS, RAZORPAY, VERCEL
E-COMMERCE & INVENTORY MANAGEMENT PLATFORM

LIVE DEMO
MAR 2026 – MAY 2026

- Architected and developed a full-stack e-commerce platform connecting parents with partner schools for class-specific uniform purchasing, featuring secure authentication, inventory management, and online payments.
- Reduced API response times by **80–85%** using **Redis** in-memory caching, significantly lowering MongoDB read load for high-traffic product catalog requests.
- Optimized database performance by up to **90%** through **MongoDB compound indexing** and aggregation pipelines, enabling faster product search, analytics, and reduced server memory usage.
- Improved frontend performance with **Next.js App Router** and image optimization, reducing JavaScript bundle size by **~40%** and image payloads by **~65%**, while strengthening application security using rate limiting, Helmet, and automated database cleanup.

HashSphere | CONSISTENT HASHING, DOCKER, REST APIs, DISTRIBUTED SYSTEMS
DISTRIBUTED KEY-VALUE STORE

SOURCE CODE 📄
AUG 2026

- Architected a distributed key-value store employing **consistent hashing** to partition and distribute data efficiently across multiple server nodes.
- Engineered a robust fault-tolerance mechanism by implementing primary-replica data replication, ensuring seamless read availability even if the primary node goes offline.
- Developed decentralized **REST API** routing logic, allowing client requests to hit any node and be automatically forwarded to the correct authoritative node.

- Containerized the multi-server network using **Docker** to simulate and test real-world distributed database scenarios, including node failures, data sync, and inter-node communication.

TECHNICAL SKILLS

Languages:	C++, JavaScript, Python, SQL
Frontend:	React.js, Flutter, Tailwind CSS, GSAP, HTML/CSS
Backend:	Node.js, Express.js, REST APIs, JWT Authentication
Databases:	MongoDB, MySQL, Redis, Firebase
Developer Tools:	Git, GitHub, Claude, Codex, Postman, VS Code, Vercel, Render
Core Competencies:	System Architecture, Performance Optimization, Concurrency Handling, API Design, Database Schema Design, Security (RBAC, Rate Limiting, Token-Based Authentication), Client & Stakeholder Management

ACHIEVEMENTS & ACTIVITIES

- Delivered 4 client projects end-to-end while independently owning development, deployment, and stakeholder communication across all engagements.
- Participated in DA-IICT Hackout and L&T Hackathon, And selected in Top 800 out of 5500+ Teams in Accenture Innovation Hackathon, building functional prototypes under time-constrained, team-based conditions.